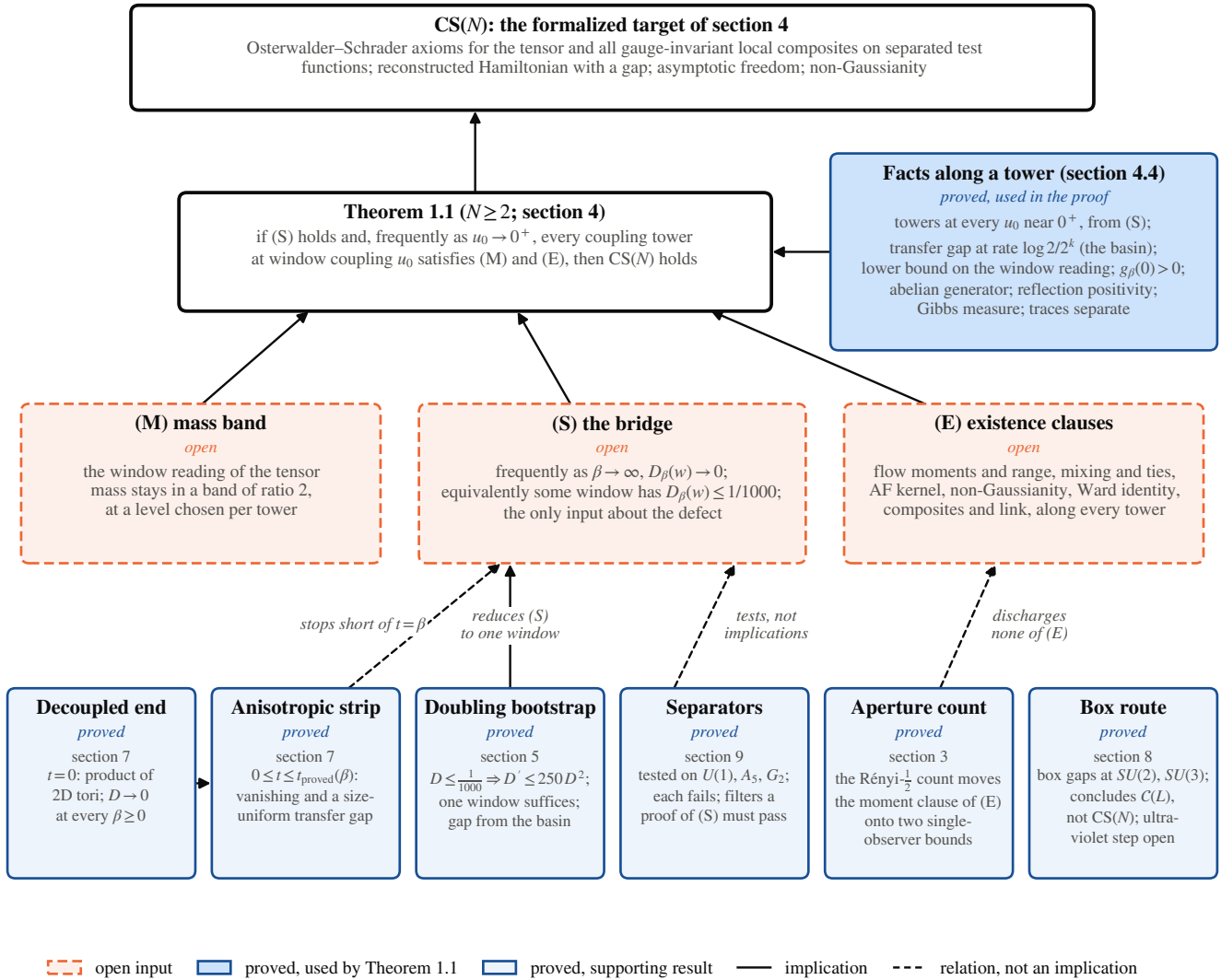
CS(N): the formalized target of section 4
Osterwalder–Schrader axioms for the tensor and all gauge-invariant local composites on separated test functions; reconstructed Hamiltonian with a gap; asymptotic freedom; non-Gaussianity
Theorem 1.1 (N ≥ 2; section 4)
if (S) holds and, frequently as u0 → 0+, every coupling tower at window coupling u0 satisfies (M) and (E), then CS(N) holds
Facts along a tower (section 4.4)
proved, used in the proof
towers at every u0 near 0+, from (S);
transfer gap at rate log 2/2^k (the basin);
lower bound on the window reading; gβ(0) > 0;
abelian generator; reflection positivity;
Gibbs measure; traces separate
(M) mass band
open
the window reading of the tensor mass stays in a band of ratio 2, at a level chosen per tower
(S) the bridge
open
frequently as β → ∞, Dβ(w) → 0;
equivalently some window has Dβ(w) ≤ 1/1000;
the only input about the defect
(E) existence clauses
open
flow moments and range, mixing and ties, AF kernel, non-Gaussianity, Ward identity, composites and link, along every tower
stops short of t = β
reduces (S) to one window
tests, not implications
discharges none of (E)
Decoupled end
proved
section 7
t = 0: product of 2D tori; D → 0 at every β ≥ 0
Anisotropic strip
proved
section 7
0 ≤ t ≤ tproved(β): vanishing and a size-uniform transfer gap
Doubling bootstrap
proved
section 5
D ≤ 1/1000 ⇒ D′ ≤ 250 D²;
one window suffices;
gap from the basin
Separators
proved
section 9
tested on U(1), A5, G2;
each fails; filters a proof of (S) must pass
Aperture count
proved
section 3
the Rényi-1/2 count moves
the moment clause of (E) onto two single-observer bounds
Box route
proved
section 8
box gaps at SU(2), SU(3);
concludes C(L),
not CS(N); ultra-violet step open
open input
proved, used by Theorem 1.1
proved, supporting result
implication
relation, not an implication